(Please write your Enrolment No. immediately) Enrolment No.

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

2nd Semester, March 2025

Paper Code: ES-102 Subject: Programming in C

Time: 1½Hrs. Max. Marks: 30

Note: Attempt Q.No.1 which is compulsory and any two more questions from remaining.

		Max. Marks	CO(s)
Q.No1	a) **Write an algorithm and draw flowchart to check whether a number is odd number or even number.**	2	CO1
	b) **How does a pre-processor work?**	2	CO1

	c) **Predict the output of the following**
	# include<stdio.h>
	void main( )
	{
	int a=4,b,c;
	b = --a;
	c = a--;
	printf("%d %d %d",a,b,c);
	return 0;
	}
		2	CO2

	d). What are basic data types supported in the C Programming Language? **e)** What is the difference between the local variable and global variable in C?	2 2	CO3
2	a) **Predict the output of the following** **# include<stdio.h>** **void main( )** { int i = 1, j = 1 ; for (; ;) { if (i > 5) break ; else j += i ; printf ("%d\n", j) ; i += j ; return 0 ; } }	5 5	CO2
	b) **Write a c Program to print Factorial of a numbers without recursion.**		
3	a) **Define an array. Write a C program to print addition of two matrices of order 3.**	5	CO2
	b) **How do u use a break and goto statement in C.**	5	CO3
4	a) **Write a program to *calculate* transpose of matrices.** b) **Write 2 different C functions to compute area and perimeter of a triangle whose sides a, b, and c are given by user as inputs.**	5 5	CO2 CO3

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Note: Attempt Q.No.1 which is compulsory and any two more questions from remaining.

		Max. Marks	CO(s)
Q.No1	a) What is an algorithm? Write an algorithm to calculate factorial of a number.	2	CO1
	b) Explain keywords & identifiers with example.	2	CO1
	c) Predict the output of the following #include<stdio.h> int main() { int i = 2, j = 3, k, l ; float a, b ; k = i / j * j ; l = j / i * i ; a = i / j * j ; b = j / i * i ; printf ("%d %d %f %f\n", k, l, a, b) ; return 0;}	2	CO2
	d) Why do we need a break and goto statement. Justify. e)What is the difference between the local variable and global variable in C.	2 2	CO3
2	a) What is an operator. Explain different types of operators used in c programming. b) Any year is input through the keyboard. Write a program to determine whether the year is a leap year or not.	5 5	CO2
3	a) Define an array. Write a Program to print the Maximum and Minimum element in an array. b) Write two different C functions to compute area and perimeter of a triangle whose sides a, b, and c are given by user as inputs. Formula to compute perimeter = a + b + c Formula to compute area = [s(s-a)(s-b)(s-c)]0.5 Where s = 0.5 * (a+b+c)	5 5	CO2 CO3
4	a) Write a program to check whether a string is a palindrome or not. b) Write a C program to check if a number has three consecutive 5s. If yes, print YES, else print NO.	5 5	CO2 CO3

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		Max. Marks	CO(s)
Q.No1	a) **Write an algorithm and draw flowchart to check whether a number is odd number or even number.**	2	CO1
	b) **How does a pre-processor work?**	2	CO1

	c) **Predict the output of the following**
	# include<stdio.h>
	void main( )
	{
	int a=4,b,c;
	b = --a;
	c = a--;
	printf("%d %d %d",a,b,c);
	return 0;
	}
		2	CO2

	d). What are basic data types supported in the C Programming Language? **e)** What is the difference between the local variable and global variable in C?	2 2	CO3
2	a) **Predict the output of the following** **# include<stdio.h>** **void main( )** { int i = 1, j = 1 ; for (; ;) { if (i > 5) break ; else j += i ; printf ("%d\n", j) ; i += j ; return 0 ; } }	5 5	CO2
	b) **Write a c Program to print Factorial of a numbers without recursion.**		
3	a) **Define an array. Write a C program to print addition of two matrices of order 3.**	5	CO2
	b) **How do u use a break and goto statement in C.**	5	CO3
4	a) **Write a program to *calculate* transpose of matrices.** b) **Write 2 different C functions to compute area and perimeter of a triangle whose sides a, b, and c are given by user as inputs.**	5 5	CO2 CO3